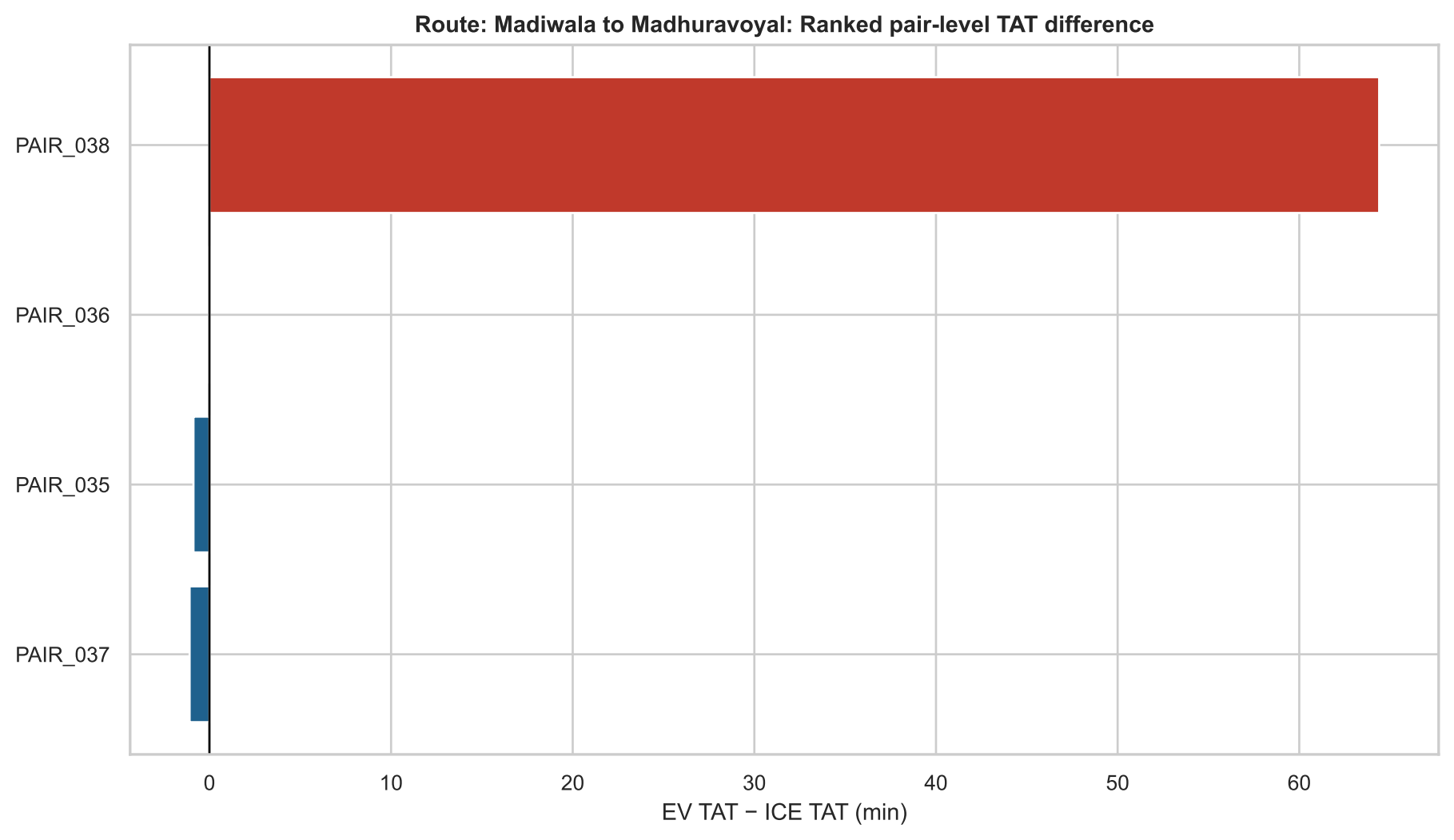
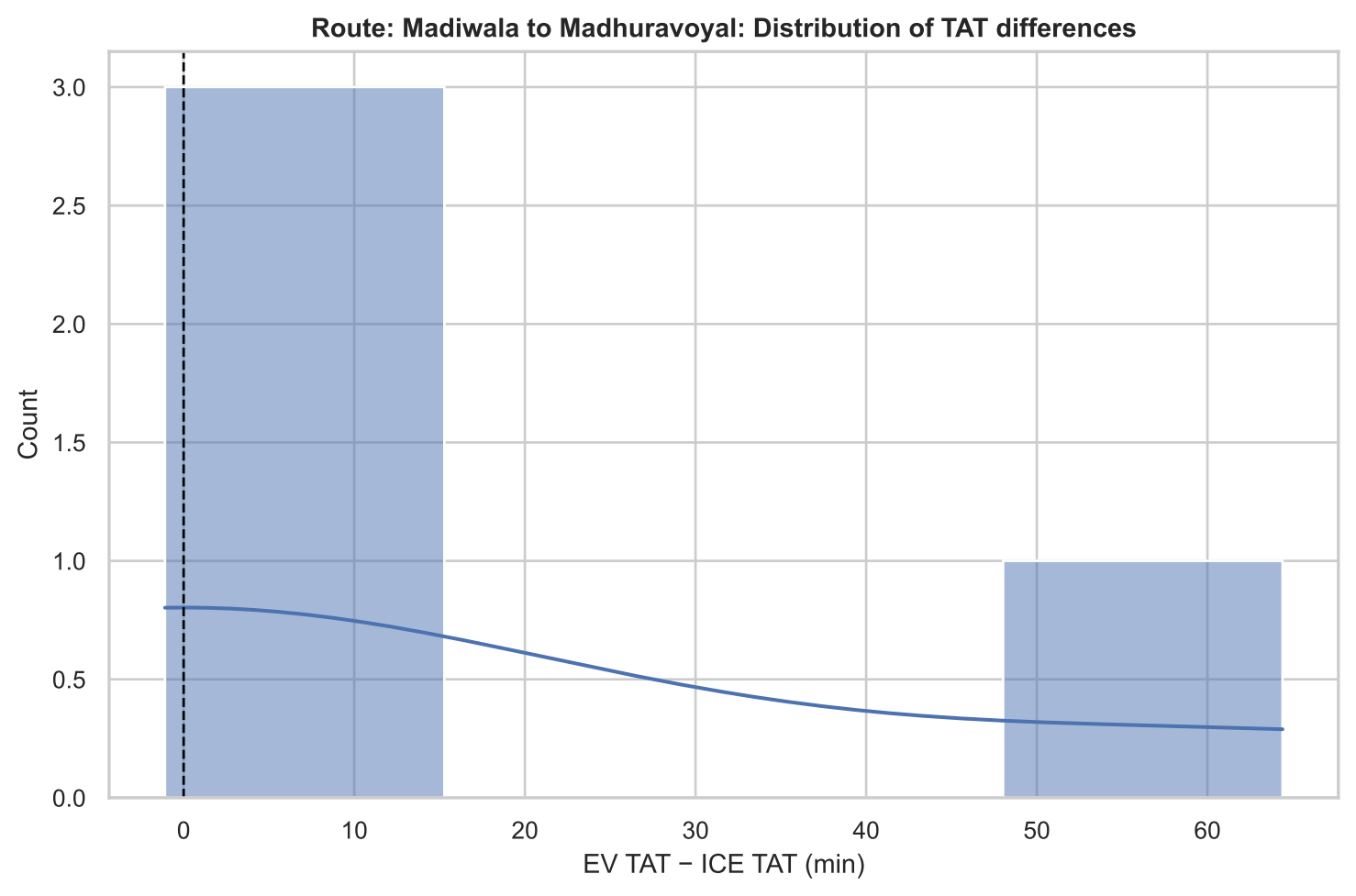
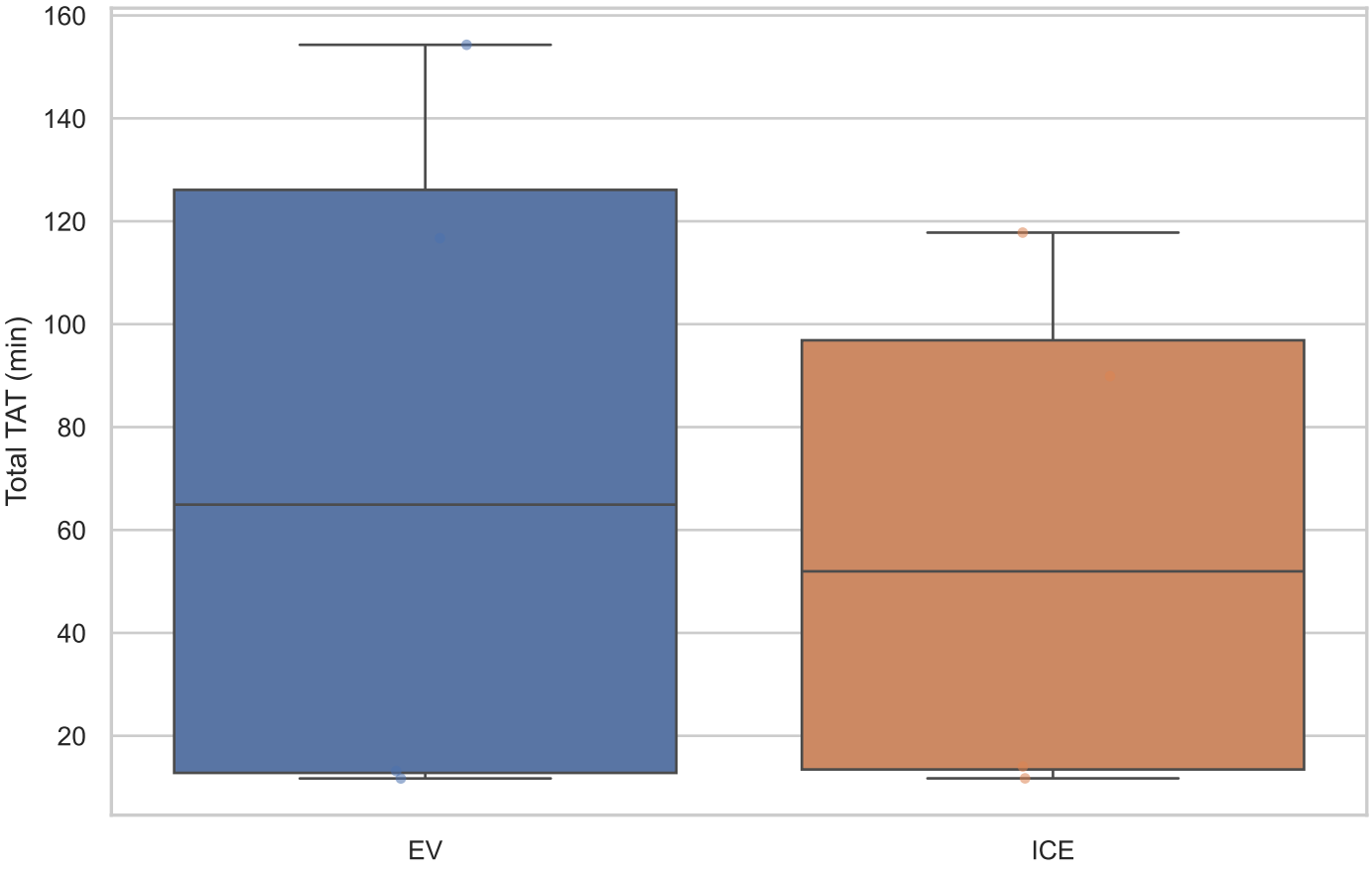
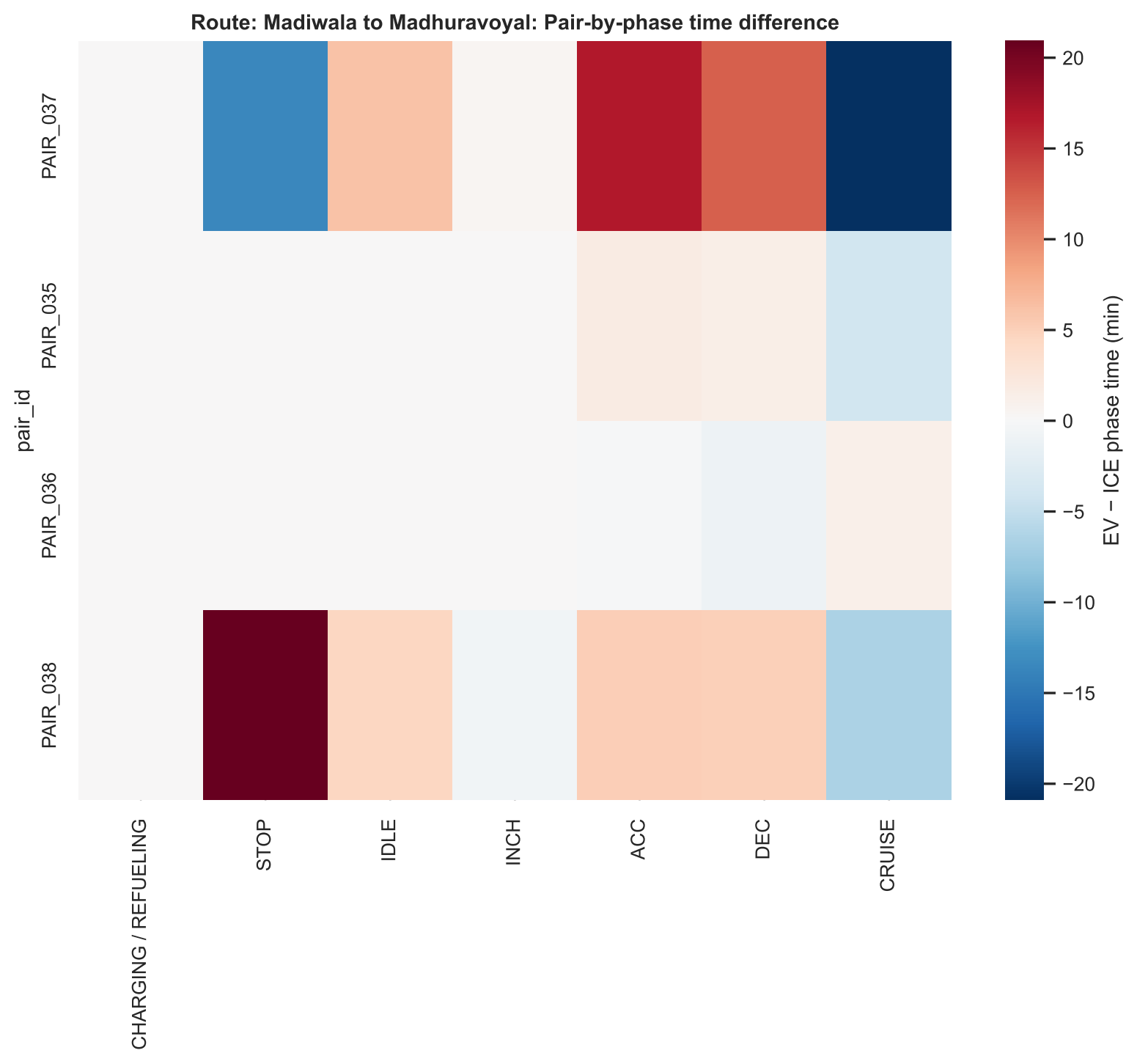
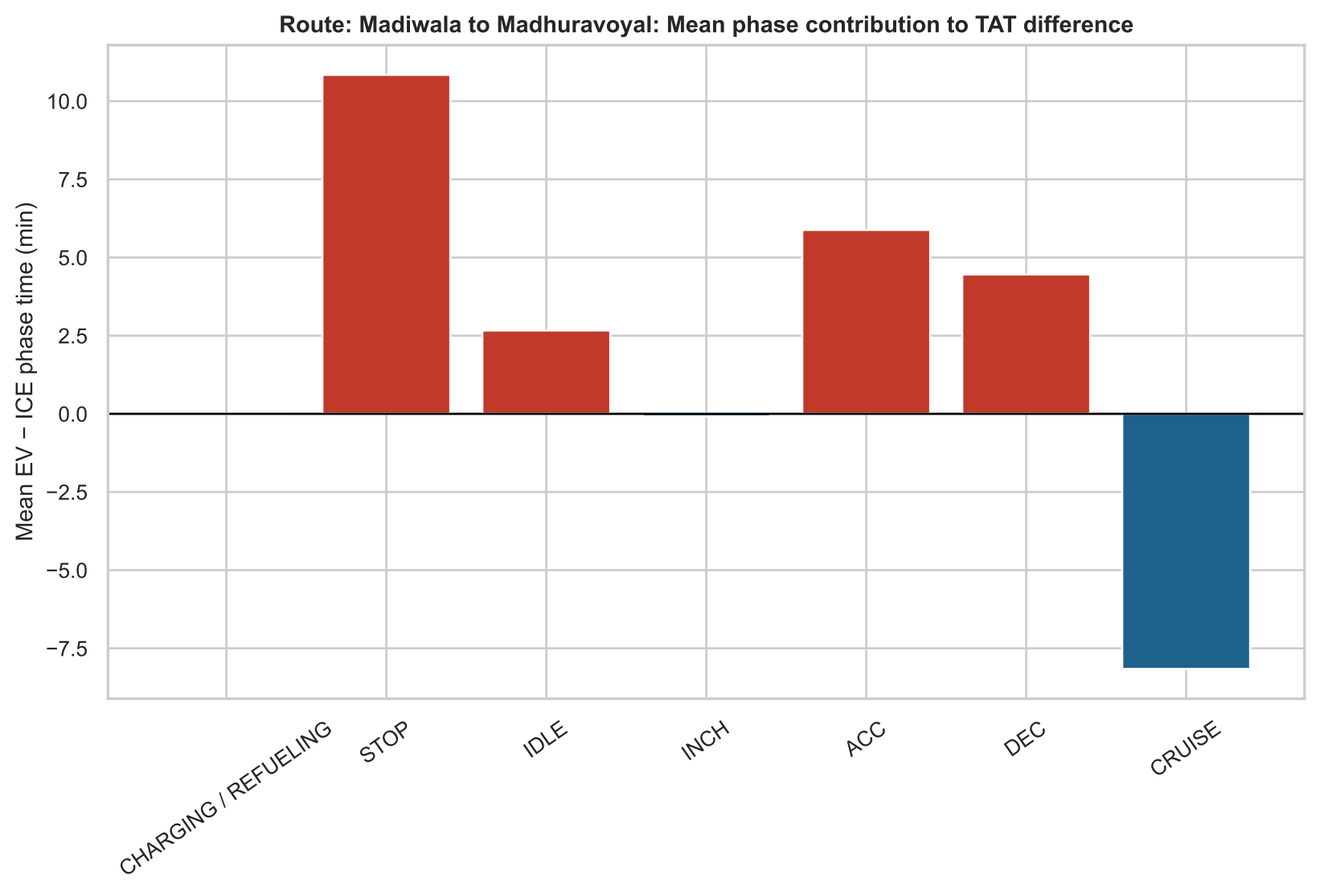
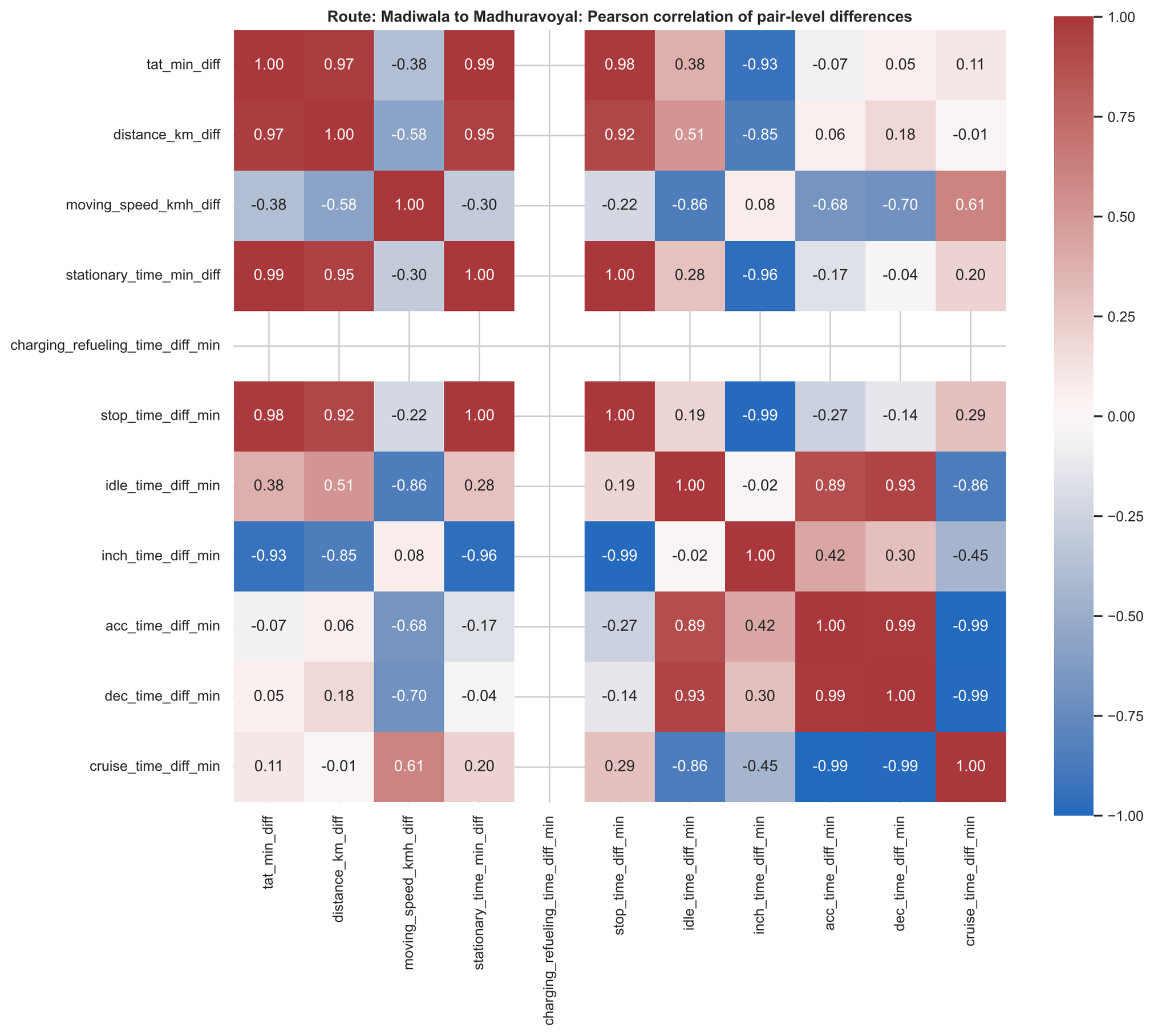

Route: Madiwala to Madhuravoyal: Ranked pair-level TAT difference
PAIR_038
PAIR_036
PAIR_035
PAIR_037
0
10
20
30
40
50
60
EV TAT − ICE TAT (min)

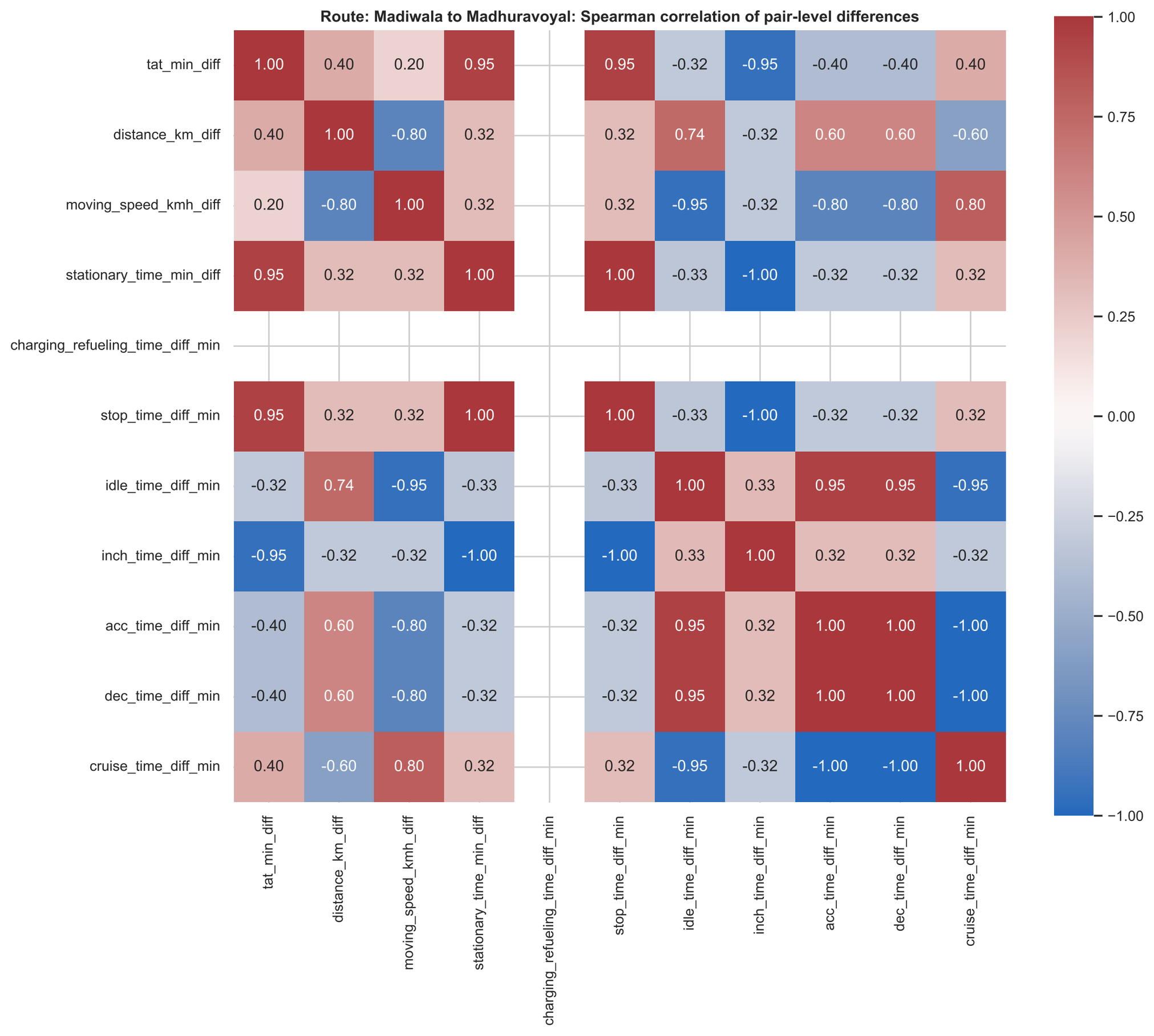

Route: Madiwala to Madhuravoyal: Distribution of TAT differences
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
50
60
EV TAT − ICE TAT (min)

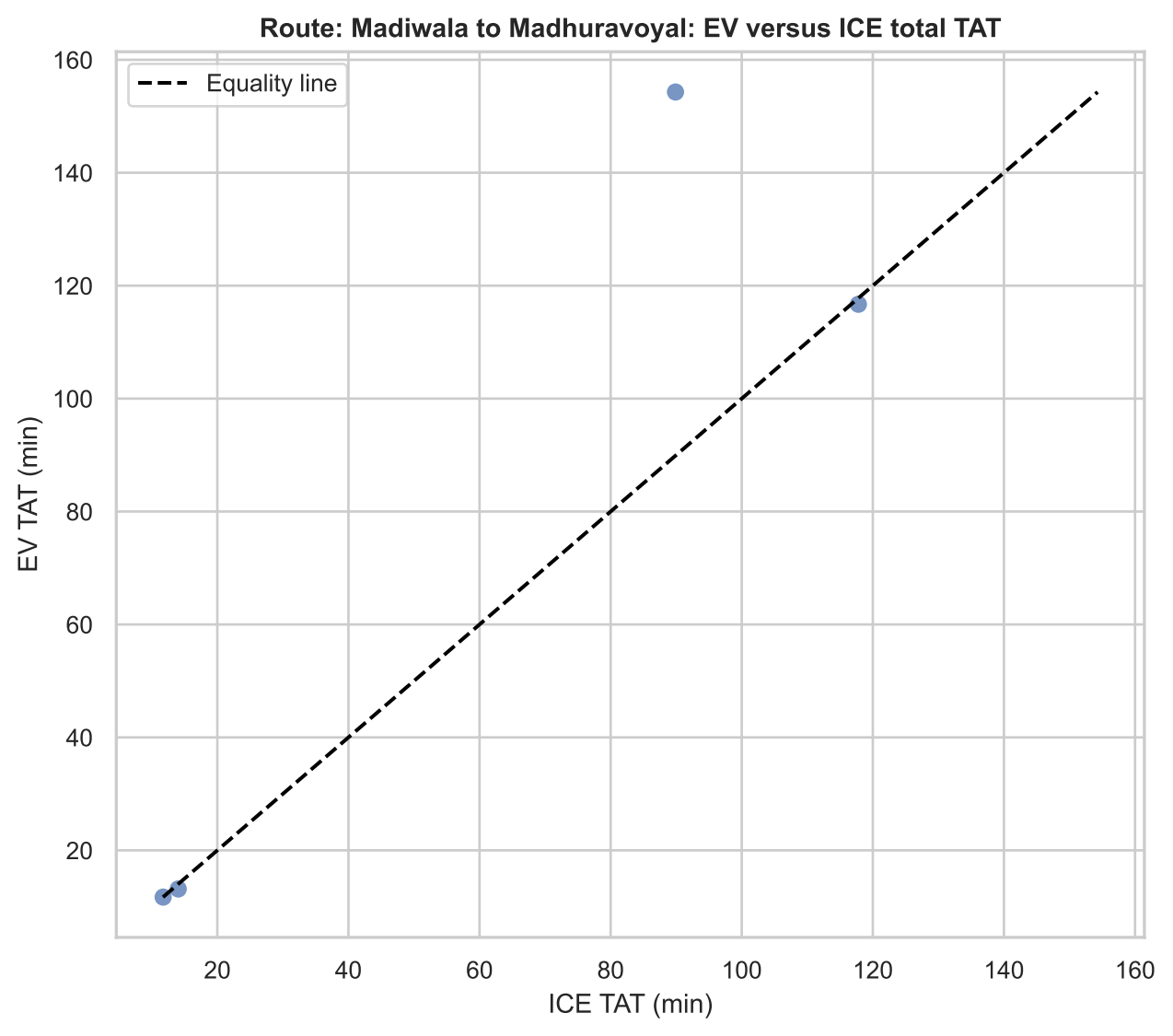

Route: Madiwala to Madhuravoyal: EV and ICE TAT distribution
Total TAT (min)
160
140
120
100
80
60
40
20
EV
ICE

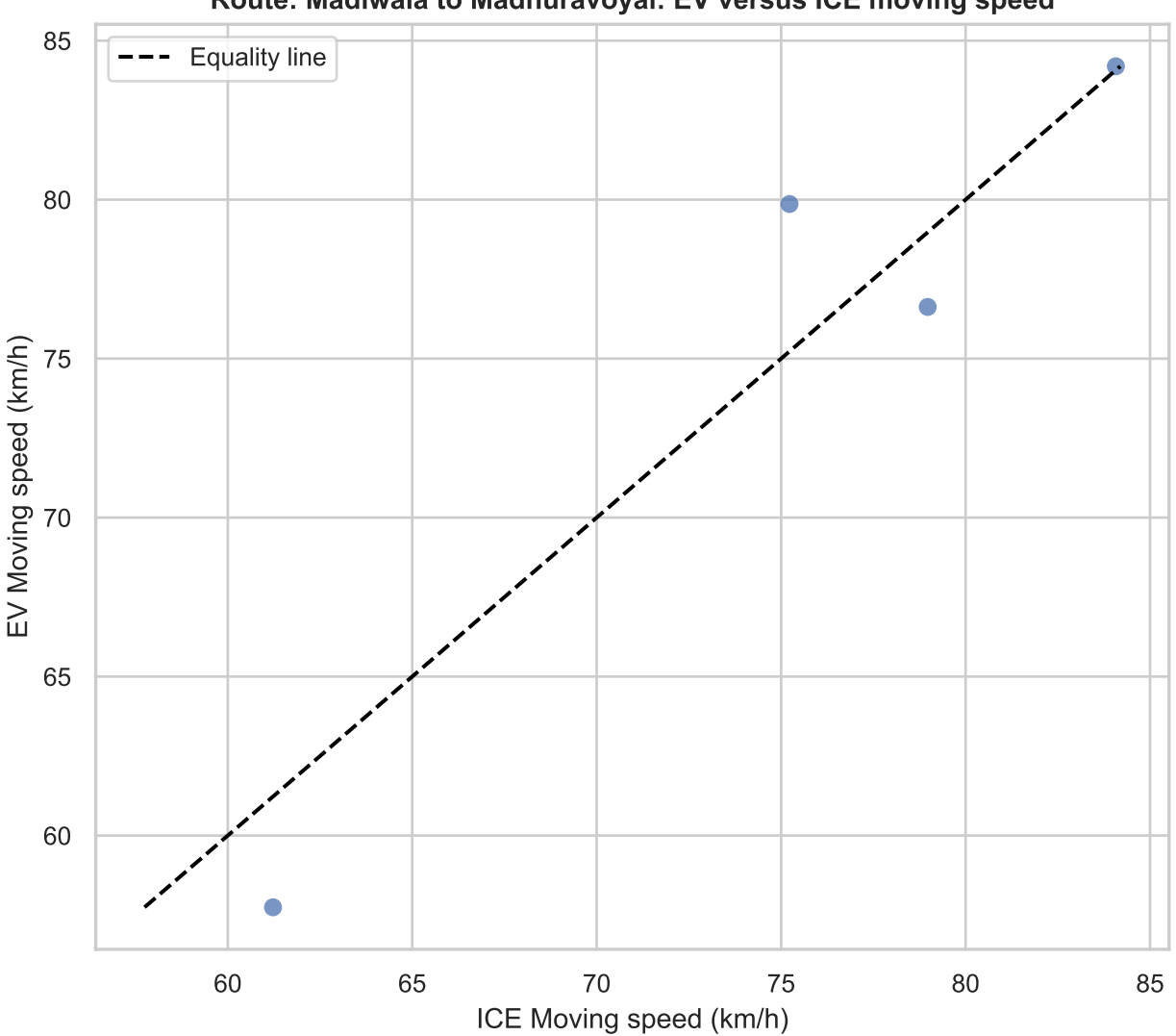

Route: Madiwala to Madhuravoyal: Pair-by-phase time difference
pair_id
PAIR_037
PAIR_035
PAIR_036
PAIR_038
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
20
15
10
5
0
−5
−10
−15
−20

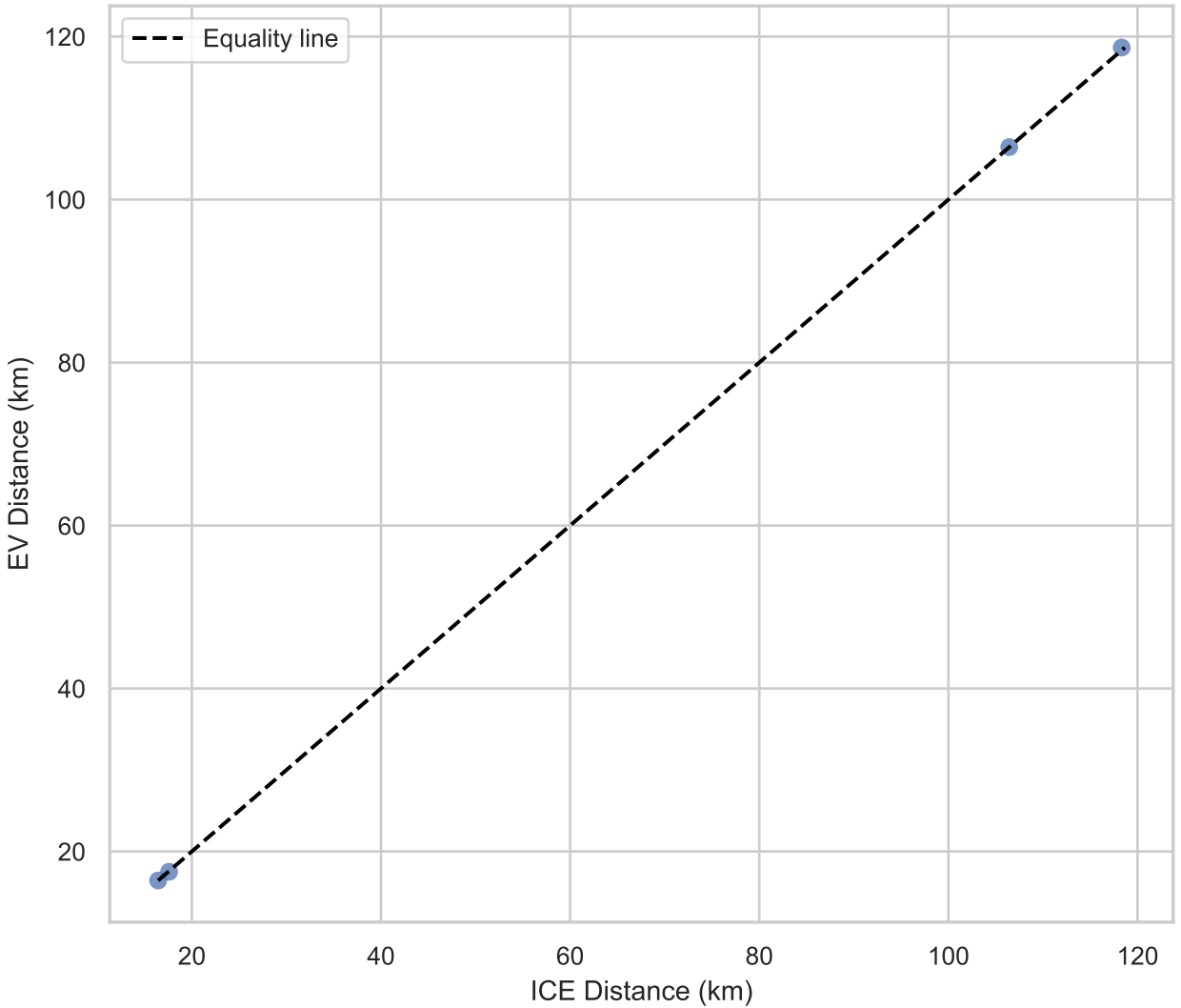

Route: Madiwala to Madhuravoyal: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

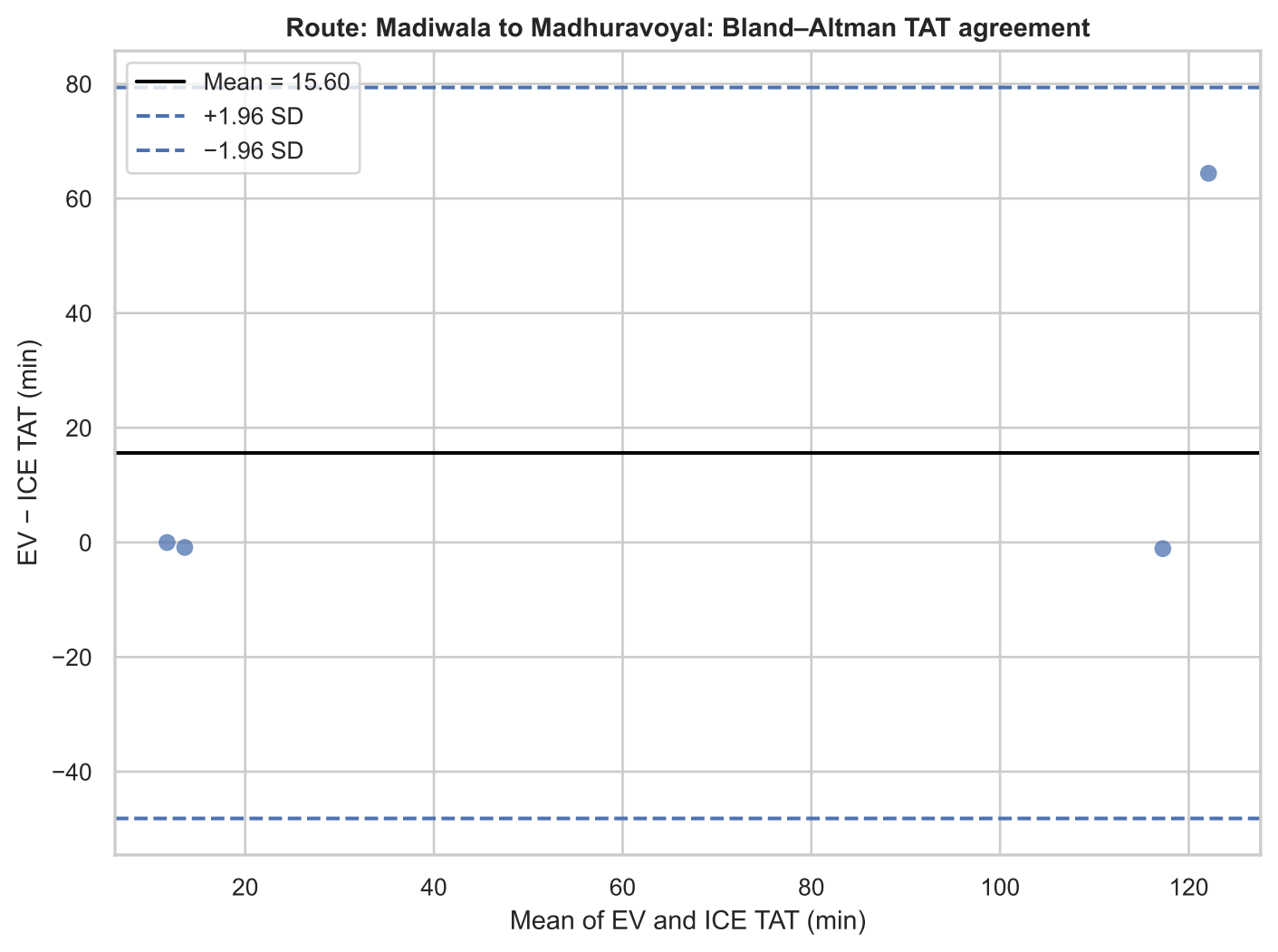

Route: Madiwala to Madhuravoyal: Pearson correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 0.97 -0.38 0.99 0.98 0.38 -0.93 -0.07 0.05 0.11
0.97 1.00 -0.58 0.95 0.92 0.51 -0.85 0.06 0.18 -0.01
-0.38 -0.58 1.00 -0.30 -0.22 -0.86 0.08 -0.68 -0.70 0.61
0.99 0.95 -0.30 1.00 1.00 0.28 -0.96 -0.17 -0.04 0.20
0.98 0.92 -0.22 1.00 1.00 0.19 -0.99 -0.27 -0.14 0.29
0.38 0.51 -0.86 0.28 0.19 1.00 -0.02 0.89 0.93 -0.86
-0.93 -0.85 0.08 -0.96 -0.99 -0.02 1.00 0.42 0.30 -0.45
-0.07 0.06 -0.68 -0.17 -0.27 0.89 0.42 1.00 0.99 -0.99
0.05 0.18 -0.70 -0.04 -0.14 0.93 0.30 0.99 1.00 -0.99
0.11 -0.01 0.61 0.20 0.29 -0.86 -0.45 -0.99 -0.99 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

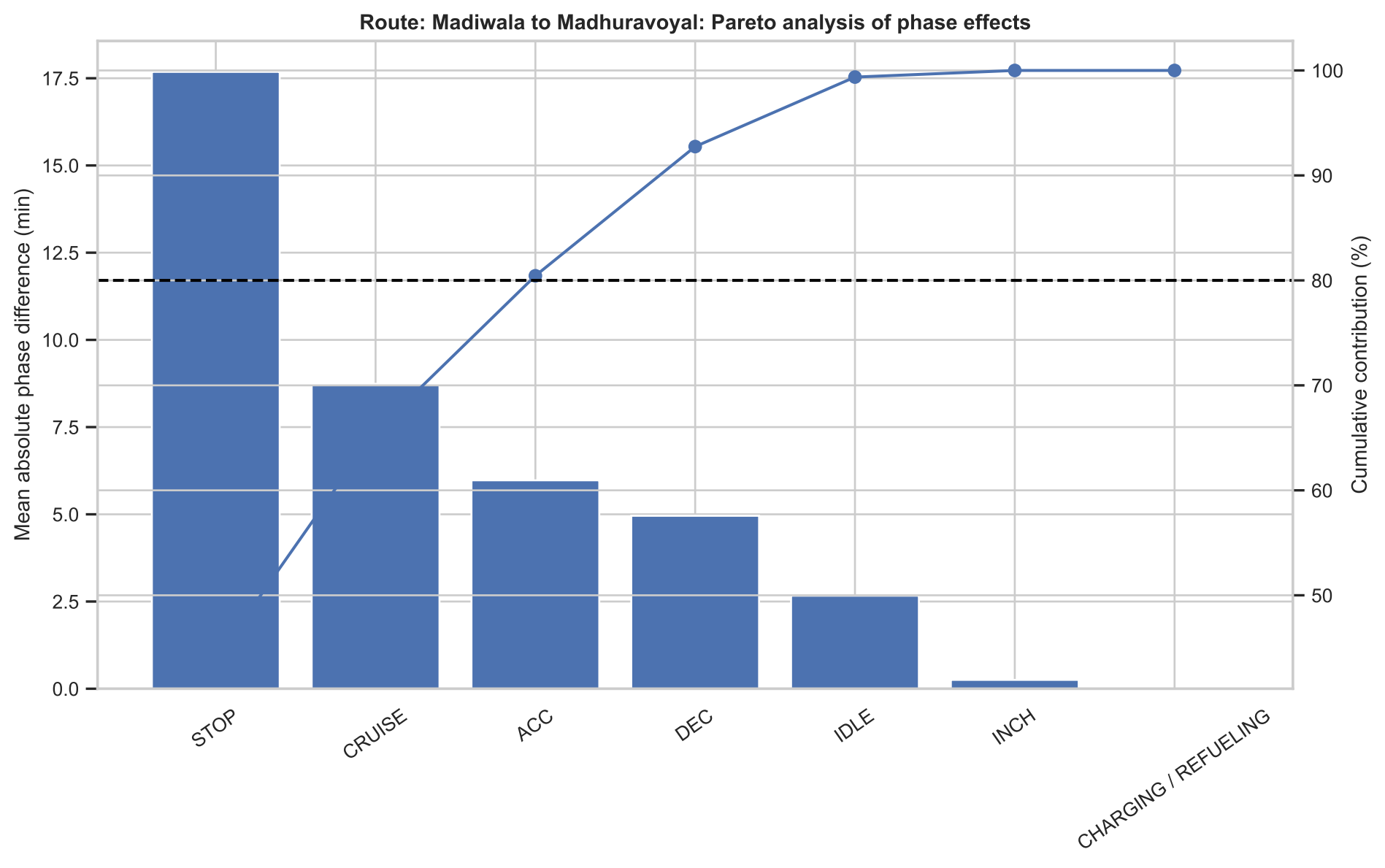

Route: Madiwala to Madhuravoyal: Spearman correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 0.40 0.20 0.95 0.95 -0.32 -0.95 -0.40 -0.40 0.40
0.40 1.00 -0.80 0.32 0.32 0.74 -0.32 0.60 0.60 -0.60
0.20 -0.80 1.00 0.32 0.32 -0.95 -0.32 -0.80 -0.80 0.80
0.95 0.32 0.32 1.00 1.00 -0.33 -1.00 -0.32 -0.32 0.32
0.95 0.32 0.32 1.00 1.00 -0.33 -1.00 -0.32 -0.32 0.32
-0.32 0.74 -0.95 -0.33 -0.33 1.00 0.33 0.95 0.95 -0.95
-0.95 -0.32 -0.32 -1.00 -1.00 0.33 1.00 0.32 0.32 -0.32
-0.40 0.60 -0.80 -0.32 -0.32 0.95 0.32 1.00 1.00 -1.00
-0.40 0.60 -0.80 -0.32 -0.32 0.95 0.32 1.00 1.00 -1.00
0.40 -0.60 0.80 0.32 0.32 -0.95 -0.32 -1.00 -1.00 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

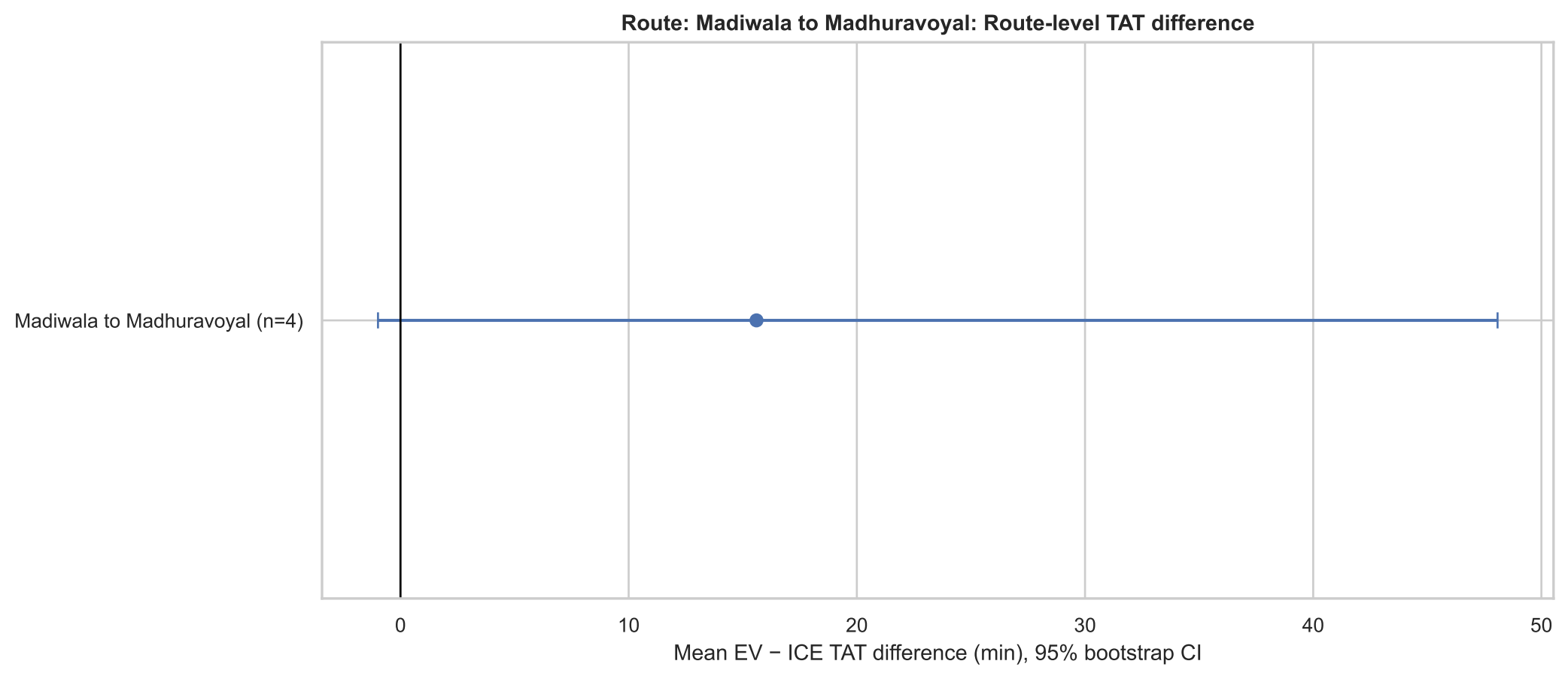

Route: Madiwala to Madhuravoyal: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
160
140
120
100
80
60
40
20
20
40
60
80
100
120
140
160

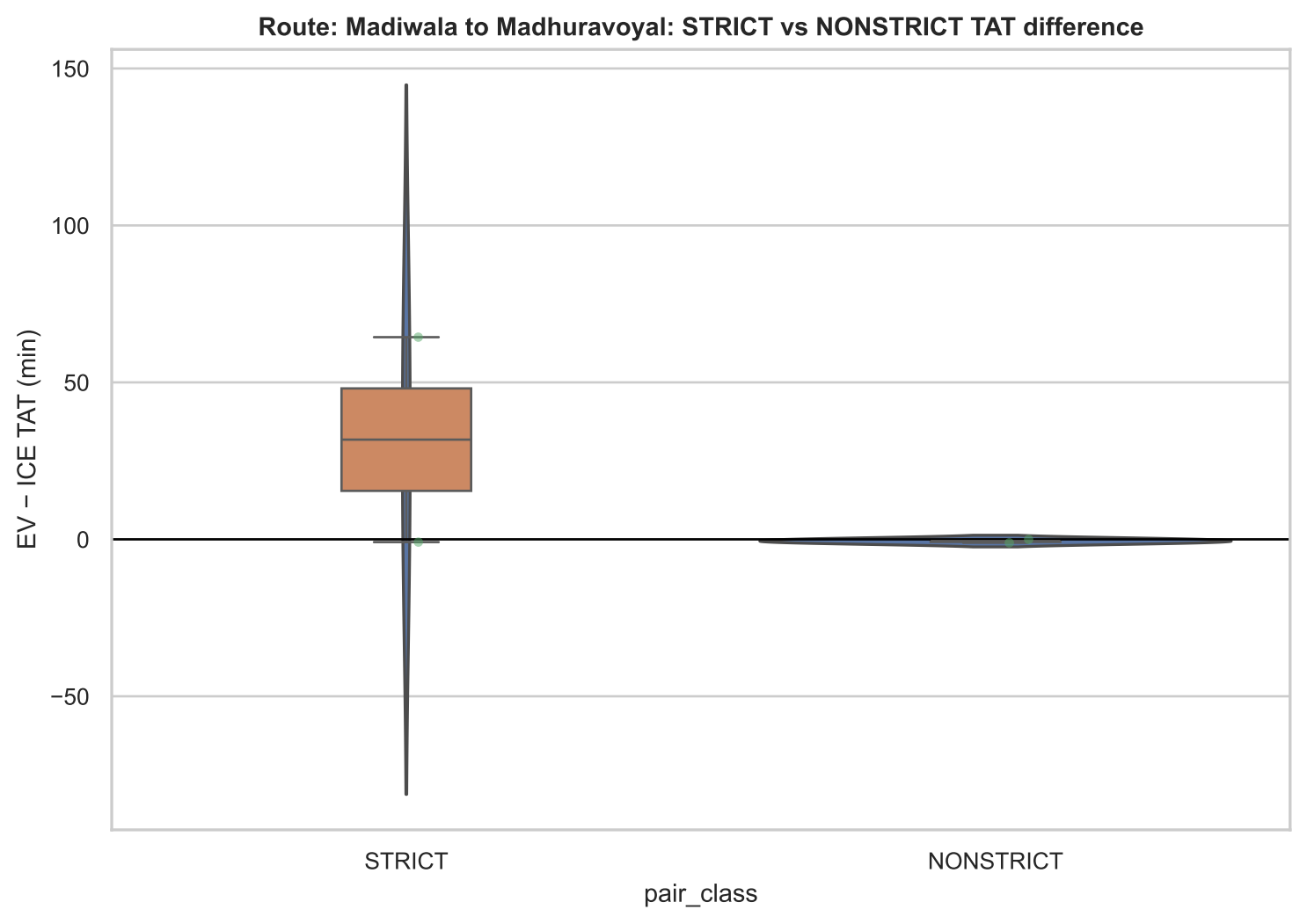

Route: Madiwala to Madhuravoyal: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
85
80
75
70
65
60
60
65
70
75
80
85

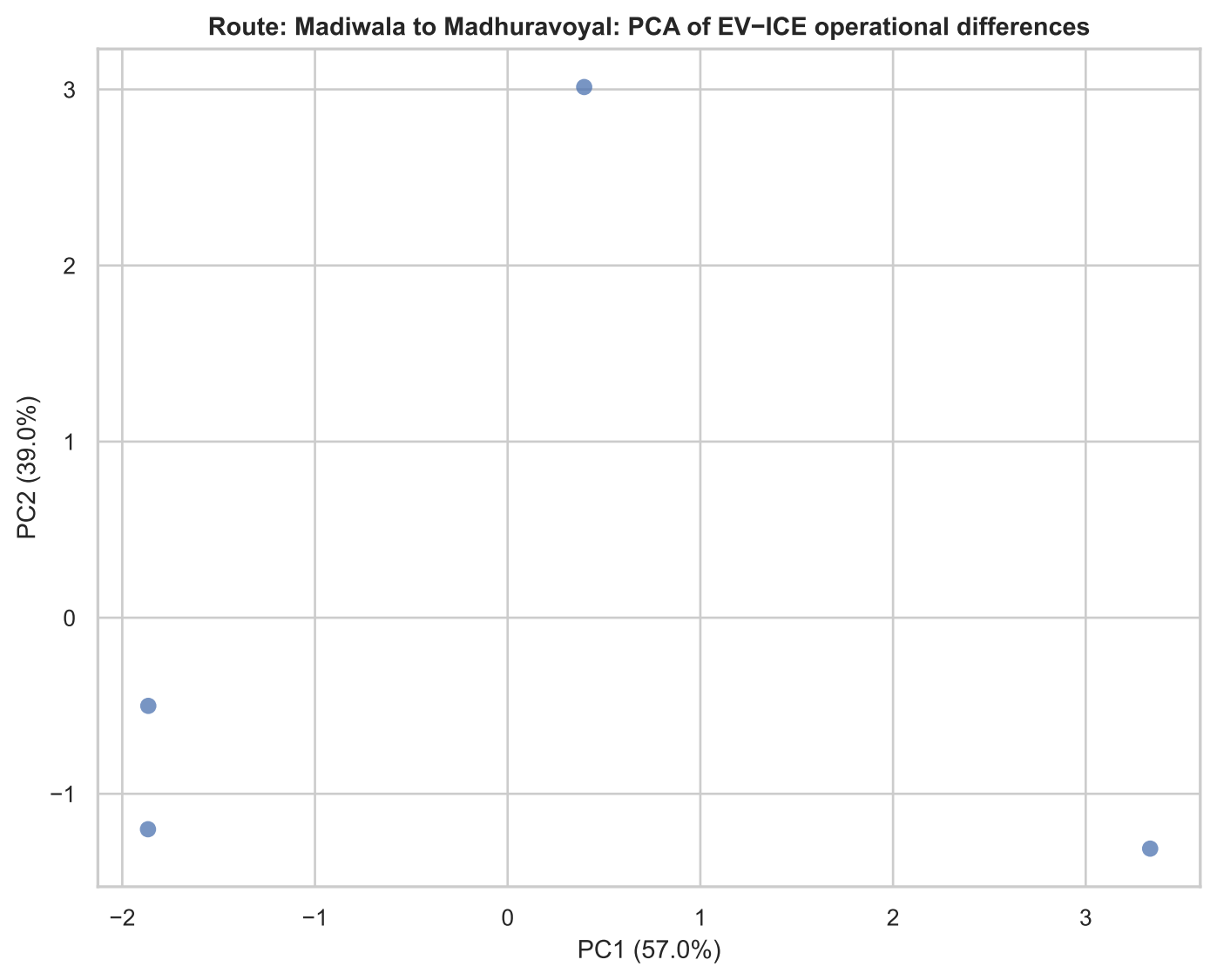

Route: Madiwala to Madhuravoyal: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
20
40
60
80
100
120

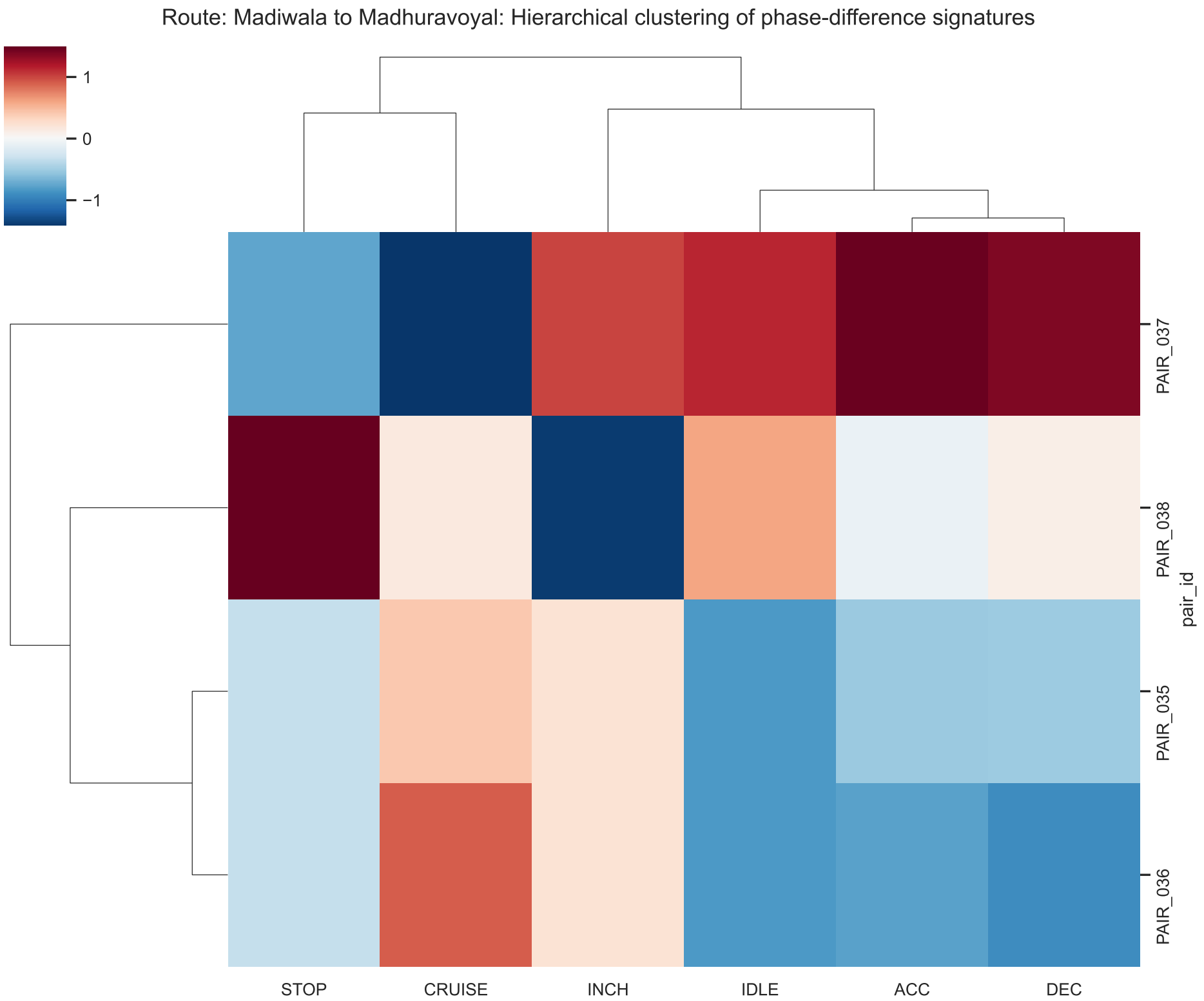

Route: Madiwala to Madhuravoyal: Bland–Altman TAT agreement
Mean = 15.60
+1.96 SD
−1.96 SD
80
60
40
20
0
−20
−40
EV − ICE TAT (min)
20
40
60
80
100
120
Mean of EV and ICE TAT (min)


Route: Madiwala to Madhuravoyal: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
50
60
70
80
90
100
STOP
CRUISE
ACC
DEC
IDLE
INCH
CHARGING / REFUELING


Route: Madiwala to Madhuravoyal: Route-level TAT difference
Madiwala to Madhuravoyal (n=4)
0
10
20
30
40
50
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Madiwala to Madhuravoyal: STRICT vs NONSTRICT TAT difference
150
100
50
0
−50
EV – ICE TAT (min)
STRICT
NONSTRICT
pair_class


Route: Madiwala to Madhuravoyal: PCA of EV−ICE operational differences
PC2 (39.0%)
PC1 (57.0%)
3
2
1
0
−1
−2
−1
0
1
2
3


Route: Madiwala to Madhuravoyal: Hierarchical clustering of phase-difference signatures
1
0
−1
PAIR_037
PAIR_038
PAIR_035
PAIR_036
pair_id
STOP
CRUISE
INCH
IDLE
ACC
DEC